# Influence of age in the opinion regarding strategies to stop global warming.

Report regarding environmental awareness by Matias Vilan

**Research Question**

To what extent does the opinion regarding strategies to reduce global warming diverge between age groups?

**Context**

Global warming is a «symptom» of climate change provoked by human's niggardly attitude towards nature. This attitude is a showcase of the profit-driven society that we have become [1] and of our short term thinking.

Furthermore, this problem is mainly caused by greenhouse gases (GHG), the most common being the CO2, that is often expelled to the atmosphere through the burning of fossil fuels. These fossil fuels are used directly or indirectly in a wide variety of activities crucial to our current society, for example to generate energy for daily use, transportation, and crucial industries. Global warming may also be caused by the methane produced largely by the livestock industry, and the aerosols (specifically fluorinated gases) found on spray cans, A.C., and Refrigerators. All these gases trap the heat in the hearth creating the Green House effect, thus creating global warming, which effects will be more visible in a not distant future, "when the planet warms, catastrophic weather and mass extinctions follow."[2]. This Apocalyptic world can be due to the biome shifts, reduction of habitats (which is already happening on the Artic circle)[3], and natural catastrophes more explored in the article *Effects of Climate Change*[4]. It is perceivable that these effects may present atrocious consequences to the future generations who will feel the toll of the errors of the past.

However, I believe that with this new generation (including my own) new measures and strategies, such as cloned meat, will be invented in order to reduce

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[1] Paddison, L. (2021). *How the rich are driving climate change*. BBC. https://www.bbc.com/future/article/20211025-climate-how-to-make-the-rich-pay-for-their-carbon-emissions

[2] Gibbens S. (2024). *How global warming is disrupting life on earth.* National Geographic. https://www.nationalgeographic.com/environment/article/global-warming-effects

[3] Younger, S. (2024). *Artic Sea Ice Near Historic Low; Antartic Ice Continues Decline.* NASA. https://www.nasa.gov/earth/arctic-sea-ice-near-historic-low-antarctic-ice-continues-decline/

[4] NASA. (n.d). *The Effects of Climate Change*. NASA science. https://science.nasa.gov/climate-change/effects/

Global Warming. Therefore, I am doing this study to understand if we have the same mentality of the people that came before us or if we are open to new approaches to this problem.

Moreover, to evaluate these divergences, I will use the Environmental Value Systems (EVSs), which are Anthropocentric, Ecocentric, and Technocentric. Hence, the EVSs are intrinsically different in their area of focus. Thus, a great way to showcase the differences between the opinions of the age groups.

**Hypothesis**

I hypothesize that young people (16-28) will be more sympathetic to the ecocentric measures, the middle aged (29-44; 45-60) more receptive to the technocentric strategies, and the old people (61-79; 80-97) more open to the anthropocentric measures.

**Variables**

Independent Variables: ages

Dependent Variables: Opinion on strategies.

**Methodology**

In order replicate this survey use the *Google Forms* tool, and inserted the title (*Strategies to Stop Global Warming*) and description as you wish, add the 6 age groups that people could be part of (16-28; 29-44; 45-60; 61-79; 80-97) and then place the statements and write the following possible answers: Strongly Agree, Agree, Disagree, and Strongly Disagree. I chose not to apply a "neutral" or "indifferent" response in order to acquire the most accurate results, as people often tend to be more neutral in several matters, for shame or fear, but this is a matter of personal discretion. The statements were inspired by two articles published on the official website of the European Commission[5] and on the United Nations website.[6] I also translated the survey to Portuguese to be more accessible to older people.

[5] https://climate.ec.europa.eu/climate-change/causes-climate-change_en
[6] https://www.un.org/en/actnow/ten-actions

**Ecocentric measures**

- Plastic products (straws, cups etc…) should be replaced by alternative materials less pollutant.
- Creating green spaces should be the government's priority.
- The use of paper and concrete should be reduced since that industries release large amounts of greenhouse gases into the environment.

**Anthropocentric measures**

- People should raise awareness among their friends and family as a measure to stop global warming.
- Individuals should implement energy saving practices in their daily routines.
- The usage of fossil fuel cars should be reduced through legislation.

**Technocentric measures**

- The use of sources of renewable energy (solar panels etc...) should be incentivized and provided by the government.
- The government should spend part of its revenue in technologies to stop global warming, such as a carbon capture industry.
- The aerosol (a fluorinate gas) in the spray cans should be replaced by air compression technology.

With the survey completed send it in the manner that best suits you. In my case I sent it via *WhatsApp* to many chat groups and relied on them to spread the survey among their own groups. I chose this method as it is the most simple and fast, and I believe that it produces the most «organic» results. When I made this survey I was aiming for 100 answers, as it is a considerably large pool sample.

The survey's strengths lie on several factors, for example, due to its simplicity it is easily replicable, the questions are specific enough to present us with precise responses, it is small and easy enough that incentivizes more people to respond as it will only take a small amount of time to answer. However, it also has its

weaknesses, such as the large options of age to choose, for it can make it difficult to analyse the data gathered.

Regarding the ethic and environmental considerations of this analyse, is accurate to state that these are minimal. The only possible factor that could occur to me is the energy spent on making the survey, researching for the study, and right this assessment. However, to minimize the environmental considerations I decreased the brightness of the screen and put the computer running in low power, in order to charge the less as possible and not contribute to the problem that I am studying. In the ethical area, I on purposely did not ask for personal information in the survey which I believe, contributes to diminish the ethical risks involved in making a survey.

**Raw Data**

| Questions / Replies Numbers | ***1*** | ***2*** | ***3*** | ***4*** | ***5*** | ***6*** | ***7*** | ***8*** | ***9*** | ***10*** |
|---|---|---|---|---|---|---|---|---|---|---|
| ***1*** | *16-28* | *SA* | *D* | *A* | *A* | *D* | *D* | *D* | *A* | *A* |
| ***2*** | *29-44* | *A* | *A* | *D* | *A* | *A* | *A* | *SA* | *A* | *SA* |
| ***3*** | *16-28* | *SA* | *A* | *A* | *A* | *D* | *A* | *A* | *A* | *A* |
| ***4*** | *16-28* | *A* | *SA* | *D* | *D* | *A* | *D* | *A* | *A* | *A* |
| ***5*** | *61-79* | *D* | *D* | *A* | *SA* | *A* | *D* | *A* | *D* | *A* |
| ***6*** | *16-28* | *SA* | *A* | *SA* | *A* | *A* | *SD* | *A* | *A* | *A* |
| ***7*** | *29-44* | *A* | *D* | *A* | *A* | *D* | *A* | *A* | *SA* | *A* |
| ***8*** | *16-28* | *SA* | *A* | *SA* | *A* | *A* | *A* | *A* | *D* | *A* |
| ***9*** | *45-60* | *A* | *SD* | *A* | *A* | *A* | *SD* | *SD* | *SD* | *A* |
| ***10*** | *16.28* | *A* | *D* | *A* | *A* | *D* | *A* | *A* | *D* | *A* |
| ***11*** | *45-60* | *A* | *A* | *D* | *A* | *A* | *A* | *SA* | *A* | *A* |
| ***12*** | *16-28* | *SA* | *A* | *A* | *D* | *A* | *D* | *D* | *D* | *A* |
| ***13*** | *29-44* | *SA* | *A* | *A* | *A* | *D* | *A* | *A* | *SA* | *A* |
| ***14*** | *16-28* | *A* | *A* | *SA* | *A* | *A* | *A* | *D* | *A* | *A* |
| ***15*** | *16-28* | *SA* | *D* | *A* | *SA* | *A* | *A* | *A* | *D* | *A* |
| ***16*** | *29-44* | *A* | *SA* | *A* | *A* | *A* | *D* | *A* | *D* | *A* |
| ***17*** | *16-28* | *SA* | *A* | *D* | *A* | *A* | *D* | *A* | *A* | *A* |
| ***18*** | *16-28* | *A* | *SA* | *A* | *D* | *A* | *A* | *A* | *SD* | *A* |
| ***19*** | *61-79* | *SA* | *A* | *A* | *A* | *SA* | *SD* | *A* | *D* | *A* |
| ***20*** | *16-28* | *SA* | *SA* | *SA* | *SA* | *A* | *A* | *A* | *SA* | *A* |
| ***21*** | *16-28* | *SA* | *A* | *SA* | *A* | *SA* | *A* | *A* | *A* | *A* |
| ***22*** | *45-60* | *A* | *SD* | *D* | *SA* | *SA* | *D* | *A* | *A* | *A* |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ***23*** | *45-60* | *A* | *SA* | *SD* | *SA* | *SD* | *A* | *A* | *A* | *A* |
| ***24*** | *16-28* | *SA* | *SA* | *A* | *D* | *A* | *A* | *A* | *D* | *A* |
| ***25*** | *16-28* | *SA* | *A* | *A* | *D* | *D* | *SA* | *A* | *A* | *A* |
| ***26*** | *16-28* | *A* | *SA* | *SA* | *SA* | *D* | *SA* | *SA* | *SA* | *A* |
| ***27*** | *29-44* | *A* | *A* | *D* | *A* | *A* | *D* | *A* | *A* | *A* |
| ***28*** | *16-28* | *SA* | *D* | *A* | *A* | *D* | *D* | *A* | *A* | *D* |
| ***29*** | *29-44* | *A* | *SD* | *D* | *A* | *A* | *A* | *SA* | *A* | *A* |
| ***30*** | *16-28* | *SA* | *SA* | *D* | *A* | *A* | *A* | *A* | *D* | *A* |
| ***31*** | *61-79* | *SA* | *A* | *D* | *SA* | *A* | *SD* | *A* | *A* | *D* |
| ***32*** | *16-28* | *SA* | *D* | *D* | *A* | *A* | *A* | *D* | *A* | *A* |
| ***33*** | *29-44* | *A* | *A* | *A* | *D* | *A* | *A* | *A* | *SA* | *A* |
| ***34*** | *29-44* | *A* | *D* | *D* | *A* | *SA* | *D* | *SA* | *SA* | *A* |
| ***35*** | *16-28* | *SA* | *SA* | *SA* | *A* | *D* | *DS* | *A* | *A* | *A* |
| ***36*** | *16-28* | *SA* | *SA* | *A* | *D* | *A* | *A* | *D* | *A* | *D* |
| ***37*** | *61-79* | *D* | *A* | *A* | *SA* | *A* | *A* | *A* | *D* | *A* |
| ***38*** | *16-28* | *SA* | *D* | *A* | *SA* | *A* | *A* | *A* | *A* | *A* |
| ***39*** | *45-60* | *A* | *A* | *A* | *SA* | *D* | *D* | *SA* | *A* | *A* |
| ***40*** | *16-28* | *A* | *D* | *SA* | *A* | *A* | *SD* | *A* | *A* | *SA* |
| ***41*** | *16-28* | *SA* | *SA* | *A* | *A* | *A* | *D* | *SA* | *SA* | *SA* |
| ***42*** | *29-44* | *A* | *A* | *A* | *D* | *A* | *SD* | *A* | *SA* | *D* |
| ***43*** | *16-28* | *SA* | *A* | *A* | *SA* | *D* | *A* | *A* | *SA* | *A* |
| ***44*** | *29-44* | *A* | *D* | *A* | *A* | *A* | *SD* | *A* | *SA* | *A* |
| ***45*** | *16-28* | *D* | *A* | *SA* | *D* | *A* | *D* | *A* | *A* | *A* |
| ***46*** | *16-28* | *SA* | *A* | *A* | *A* | *A* | *SD* | *D* | *A* | *A* |
| ***47*** | *80-97* | *A* | *SA* | *D* | *A* | *A* | *D* | *A* | *A* | *D* |
| ***48*** | *16-28* | *A* | *A* | *D* | *A* | *A* | *A* | *SA* | *D* | *A* |
| ***49*** | *29-44* | *SA* | *A* | *A* | *D* | *A* | *D* | *A* | *SA* | *A* |
| ***50*** | *16-28* | *SA* | *A* | *A* | *D* | *A* | *A* | *A* | *SA* | *D* |
| ***51*** | *16-28* | *SA* | *D* | *A* | *A* | *SA* | *A* | *D* | *SA* | *A* |
| ***52*** | *16-28* | *SA* | *A* | *A* | *A* | *D* | *A* | *A* | *D* | *D* |
| ***53*** | *16-28* | *A* | *D* | *A* | *D* | *SA* | *D* | *A* | *A* | *A* |
| ***54*** | *29-44* | *A* | *A* | *A* | *A* | *A* | *A* | *SA* | *A* | *A* |
| ***55*** | *16-28* | *SA* | *A* | *D* | *A* | *D* | *SD* | *D* | *A* | *D* |
| ***56*** | *16-28* | *A* | *SA* | *A* | *D* | *A* | *D* | *A* | *D* | *A* |
| ***57*** | *16-28* | *SA* | *A* | *A* | *D* | *A* | *SD* | *D* | *D* | *A* |
| ***58*** | *29-44* | *SA* | *SD* | *A* | *A* | *A* | *A* | *D* | *A* | *D* |
| ***59*** | *16-28* | *A* | *A* | *SA* | *D* | *A* | *D* | *D* | *A* | *A* |
| ***60*** | *16-28* | *SA* | *D* | *A* | *A* | *A* | *D* | *A* | *SA* | *D* |
| ***61*** | *29-44* | *D* | *D* | *A* | *A* | *A* | *A* | *SA* | *A* | *D* |
| ***62*** | *16-28* | *A* | *A* | *D* | *A* | *D* | *D* | *A* | *D* | *A* |
| ***63*** | *16-28* | *SA* | *A* | *D* | *D* | *A* | *A* | *D* | *A* | *D* |
| ***64*** | *16-28* | *A* | *A* | *SA* | *D* | *A* | *SD* | *D* | *A* | *D* |
| ***65*** | *61-79* | *D* | *A* | *D* | *A* | *SA* | *D* | *A* | *A* | *A* |
| ***66*** | *16-28* | *SA* | *D* | *D* | *A* | *A* | *A* | *D* | *A* | *A* |
| ***67*** | *16-28* | *SA* | *A* | *A* | *A* | *D* | *A* | *A* | *D* | *D* |
| ***68*** | *45-60* | *A* | *A* | *D* | *A* | *D* | *SD* | *SA* | *A* | *A* |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ***69*** | *16-28* | *SA* | *SA* | *A* | *D* | *A* | *D* | *D* | *D* | *A* |
| ***70*** | *16-28* | *A* | *D* | *SA* | *D* | *A* | *A* | *D* | *A* | *A* |
| ***71*** | *29-44* | *A* | *A* | *D* | *A* | *D* | *A* | *A* | *SA* | *D* |
| ***72*** | *16-28* | *SA* | *D* | *A* | *A* | *D* | *SD* | *A* | *D* | *A* |
| ***73*** | *16-28* | *SA* | *A* | *SA* | *D* | *A* | *A* | *D* | *SA* | *A* |
| ***74*** | *45-60* | *A* | *A* | *D* | *A* | *A* | *A* | *A* | *A* | *D* |
| ***75*** | *16-28* | *SA* | *SD* | *A* | *D* | *A* | *D* | *SD* | *D* | *A* |
| ***76*** | *16-28* | *A* | *D* | *A* | *A* | *D* | *D* | *A* | *A* | *A* |
| ***77*** | *29-44* | *D* | *A* | *SA* | *A* | *A* | *D* | *D* | *A* | *A* |
| ***78*** | *16-28* | *SA* | *D* | *A* | *D* | *A* | *A* | *A* | *A* | *D* |
| ***79*** | *29-44* | *A* | *A* | *SA* | *A* | *D* | *D* | *A* | *A* | *A* |
| ***80*** | *16-28* | *SA* | *D* | *A* | *D* | *A* | *SD* | *D* | *A* | *A* |
| ***81*** | *16-28* | *A* | *SA* | *D* | *A* | *D* | *A* | *A* | *A* | *D* |
| ***82*** | *29-44* | *SA* | *A* | *A* | *D* | *SA* | *SD* | *D* | *A* | *A* |
| ***83*** | *16-28* | *A* | *A* | *D* | *SA* | *SA* | *D* | *A* | *D* | *A* |
| ***84*** | *16-28* | *SA* | *A* | *SA* | *A* | *A* | *D* | *A* | *D* | *D* |
| ***85*** | *16-28* | *D* | *A* | *D* | *A* | *SA* | *D* | *A* | *D* | *D* |
| ***86*** | *61-79* | *A* | *A* | *D* | *A* | *A* | *SA* | *A* | *A* | *A* |
| ***87*** | *29-44* | *A* | *SD* | *A* | *A* | *D* | *D* | *A* | *A* | *SA* |
| ***88*** | *16-28* | *A* | *D* | *A* | *D* | *SA* | *SD* | *D* | *D* | *A* |
| ***89*** | *16-28* | *SA* | *A* | *A* | *D* | *A* | *SA* | *A* | *A* | *A* |
| ***90*** | *16-28* | *SA* | *A* | *A* | *SD* | *A* | *SD* | *SA* | *SD* | *A* |
| ***91*** | *45-60* | *A* | *SA* | *A* | *A* | *D* | *A* | *A* | *A* | *SA* |
| ***92*** | *16-28* | *A* | *SA* | *D* | *A* | *D* | *A* | *SA* | *SA* | *A* |

*Table 1 Answers to the Survey*

My survey only received 92 answers, not meeting my target of 110 answers. However, the difference is small, hence not affecting the results of the investigation. From the 92 responses, 58 chose the age group 16-28, 19 chose 29-44, 9 chose 45-60, 5 the 61-79 and 1 chose the option 80-97. Given the great sample size of the age group 16-28 (naturally, due to the greater youth's familiarity with the technology), the results will be very reliable. In contrast due to the opposite effect the results of the other age groups (except 29-44 that has a considerable amount of responses) are lower making them less reliable, in order to solve this I should had used another method for this age groups.

**Processed Data**

| Replies Number | ***Ages*** | ***Ecocentric Questions*** | ***Anthropocentric Questions*** | ***Technocentric Questions*** | ***EVS*** |
|---|---|---|---|---|---|
| ***1*** | *16-28* | *3* | *2.3* | *2.67* | *Ecocentric* |
| ***2*** | *29-44* | *2.67* | *3* | *3.67* | *Technocentric* |
| ***3*** | *16-28* | *3.33* | *2.33* | *3* | *Ecocentric* |
| ***4*** | *16-28* | *3* | *2.33* | *3* | *Ecocentric* |
| ***5*** | *61-79* | *2.33* | *2.67* | *3.3* | *Technocentric* |
| ***6*** | *16-28* | *3.66* | *2.33* | *3* | *Ecocentric* |
| ***7*** | *29-44* | *2.66* | *2.66* | *3.33* | *Technocentric* |
| ***8*** | *16-28* | *3* | *3* | *2.66* | *Anthro/Eco* |
| ***9*** | *16-28* | *2.67* | *2.33* | *2.33* | *Ecocentric* |
| ***10*** | *16.28* | *3* | *2.67* | *2.67* | *Ecocentric* |
| ***11*** | *45-60* | *2.67* | *3* | *3.33* | *Technocentric* |
| ***12*** | *16-28* | *3.33* | *2.33* | *2.33* | *Ecocentric* |
| ***13*** | *29-44* | *3* | *3.33* | *2.67* | *Anthropocentric* |
| ***14*** | *16-28* | *3.33* | *3* | *2.67* | *Ecocentric* |
| ***15*** | *16-28* | *3* | *3.33* | *2.67* | *Anthropocentric* |
| ***16*** | *29-44* | *3.33* | *2.67* | *2.67* | *Ecocentric* |
| ***17*** | *16-28* | *3* | *2.67* | *3* | *Ecocentric* |
| ***18*** | *16-28* | *3.33* | *2.33* | *2.33* | *Ecocentric* |
| ***19*** | *61-79* | *3.33* | *2.67* | *2.67* | *Ecocentric* |
| ***20*** | *16-28* | *4* | *3.33* | *3.33* | *Ecocentric* |
| ***21*** | *16-28* | *3.67* | *3.33* | *3* | *Ecocentric* |
| ***22*** | *45-60* | *2* | *3.33* | *3* | *Anthropocentric* |
| ***23*** | *45-60* | *2.67* | *2.67* | *3* | *Technocentric* |
| ***24*** | *16-28* | *3.67* | *2.67* | *2.67* | *Ecocentric* |
| ***25*** | *16-28* | *3.33* | *2.67* | *3* | *Ecocentric* |
| ***26*** | *16-28* | *3.67* | *3.33* | *3.67* | *Ecocentric* |
| ***27*** | *29-44* | *2.67* | *2.67* | *3* | *Technocentric* |
| ***28*** | *16-28* | *3* | *2.33* | *2.66* | *Ecocentric* |
| ***29*** | *29-44* | *3* | *3* | *3.33* | *Technocentric* |
| ***30*** | *16-28* | *3.67* | *3* | *2.67* | *Ecocentric* |
| ***31*** | *61-79* | *3* | *2.67* | *2.67* | *Ecocentric* |
| ***32*** | *16-28* | *2.67* | *3* | *2.67* | *Anthropocentric* |
| ***33*** | *29-44* | *3* | *2.67* | *3.33* | *Technocentric* |
| ***34*** | *29-44* | *2.33* | *3* | *3.67* | *Technocentric* |
| ***35*** | *16-28* | *4* | *2* | *3* | *Ecocentric* |

| | | | | | |
|---|---|---|---|---|---|
| ***36*** | *16-28* | *3.67* | *2.67* | *2.33* | *Ecocentric* |
| ***37*** | *61-79* | *2.67* | 3.33 | 2.67 | *Ecocentric* |
| ***38*** | *16-28* | *3* | *3.33* | *3* | *Anthropocentric* |
| ***39*** | *45-60* | *3* | *2.67* | *3.33* | *Technocentric* |
| ***40*** | *16-28* | *3* | *2.33* | *3.33* | *Technocentric* |
| ***41*** | *16-28* | *3.67* | *2.67* | *4* | *Technocentric* |
| ***42*** | *29-44* | *3* | *2* | *3* | *Ecocentric* |
| ***43*** | *16-28* | *3.33* | *3* | *3.33* | *Ecocentric* |
| ***44*** | *29-44* | *2.67* | *2.33* | *3.33* | *Technocentric* |
| ***45*** | *16-28* | *3* | *2.33* | *3* | *Ecocentric* |
| ***46*** | *16-28* | *3.33* | *2.33* | *2.67* | *Ecocentric* |
| ***47*** | *80-97* | *3* | *2.67* | *2.67* | *Ecocentric* |
| ***48*** | *16-28* | *2.67* | *3* | *3* | *Anthropocentric* |
| ***49*** | *29-44* | *3.33* | *2.33* | *3.33* | *Ecocentric* |
| ***50*** | *16-28* | *3.33* | *2.67* | *3* | *Ecocentric* |
| ***51*** | *16-28* | *3* | *3.33* | *3* | *Anthropocentric* |
| ***52*** | *16-28* | *3.33* | *2.67* | *2.33* | *Ecocentric* |
| ***53*** | *16-28* | *2.67* | *2.67* | *3* | *Technocentric* |
| ***54*** | *29-44* | *3* | *3* | *3.33* | *Technocentric* |
| ***55*** | *16-28* | *3* | *2* | *2.33* | *Ecocentric* |
| ***56*** | *16-28* | *3.33* | *2.33* | *2.67* | *Ecocentric* |
| ***57*** | *16-28* | *3.33* | *2* | *2.33* | *Ecocentric* |
| ***58*** | *29-44* | *2.67* | *3* | *2.33* | *Anthropocentric* |
| ***59*** | *16-28* | *3.33* | *2.33* | *2.67* | *Ecocentric* |
| ***60*** | *16-28* | *3* | *2.67* | *3* | *Ecocentric* |
| ***61*** | *29-44* | *2.33* | *3* | *3* | *Anthropocentric* |
| ***62*** | *16-28* | *2.67* | *2.33* | *2.67* | *Ecocentric* |
| ***63*** | *16-28* | *3* | *2.67* | *2.33* | *Ecocentric* |
| ***64*** | *16-28* | *3.33* | *2* | *2.33* | *Ecocentric* |
| ***65*** | *61-79* | *2.33* | *3* | *3* | *Anthropocentric* |
| ***66*** | *16-28* | *2.67* | *3* | *2.67* | *Anthropocentric* |
| ***67*** | *16-28* | *3.33* | *2.67* | *2,33* | *Ecocentric* |
| ***68*** | *45-60* | *2.67* | *2* | *3.33* | *Technocentric* |
| ***69*** | *16-28* | *3.67* | *2.33* | *2.33* | *Ecocentric* |
| ***70*** | *16-28* | *3.33* | *2.67* | *3* | *Ecocentric* |
| ***71*** | *29-44* | *2.67* | *2.67* | *3.67* | *Technocentric* |
| ***72*** | *16-28* | *3* | *2* | *2.33* | *Ecocentric* |
| ***73*** | *16-28* | *3.67* | *2.67* | *3* | *Ecocentric* |
| ***74*** | *45-60* | *2.67* | *3* | *2.67* | *Anthropocentric* |

| | | | | | |
|---|---|---|---|---|---|
| ***75*** | *16-28* | *2.67* | *2.33* | *2* | *Ecocentric* |
| ***76*** | *16-28* | *2.67* | *2.33* | *3* | *Technocentric* |
| ***77*** | *29-44* | *3* | *2.67* | *2.67* | *Ecocentric* |
| ***78*** | *16-28* | *3* | *2.33* | *3* | *Ecocentric* |
| ***79*** | *29-44* | *3.33* | *2.33* | *3* | *Ecocentric* |
| ***80*** | *16-28* | *3* | *2* | *2.67* | *Ecocentric* |
| ***81*** | *16-28* | *3* | *2.67* | *2.67* | *Ecocentric* |
| ***82*** | *29-44* | *3.33* | *2.33* | *2.67* | *Ecocentric* |
| ***83*** | *16-28* | *2.67* | *3.33* | *2.67* | *Anthropocentric* |
| ***84*** | *16-28* | *3.67* | *2.67* | *2.33* | *Ecocentric* |
| ***85*** | *16-28* | *2.33* | *3* | *2.33* | *Anthropocentric* |
| ***86*** | *61-79* | *2.67* | *3.33* | *3* | *Anthropocentric* |
| ***87*** | *29-44* | *2.33* | *2.33* | *3.33* | *Technocentric* |
| ***88*** | *16-28* | *2.67* | *2.33* | *2.33* | *Ecocentric* |
| ***89*** | *16-28* | *3.33* | *3* | *3* | *Ecocentric* |
| ***90*** | *16-28* | *3.33* | *1.67* | *2.67* | *Ecocentric* |
| ***91*** | *45-60* | *3.33* | *2.67* | *3.33* | *Ecocentric* |
| ***92*** | *16-28* | *3* | *2.67* | *3.67* | *Technocentric* |

*Table 2 EVSs by age group*

To calculate the EVSs, I attributed a score to each option that a person could choose (SA-4; A-3; D-2; SD-1), then added the three options selected and then divided by three (the number of questions in each EVSs set), then I compared the score of each EVSs question group, and the one with the highest score was the person's EVSs. Although this method has its benefits, it is simple to use, for instance, it is not the best and it can be unprecise, for example some of the EVSs groups had the same score which really difficulted the analysis (8, 26, 43, 48, 60, 62, 65, 79, 92), in this cases, for instance the group of ecocentric questions had the same score as the technocentric one, I would simply "prioritize" the ecocentric one has the responder can be ecocentric while believing that the current technology can play its role in stopping global warming. I used the same criteria when, instead of the ecocentric, the anthropocentric had the same value as the technocentric.

In order to see if age can, in fact, influence the opinions regarding the strategies to stop global warming, I used Pearson Test, since it is simple, fast, and more importantly reliable. To calculate the *r (*indicates the direction and the strength of

the correlation) I had to modify the x variable, the age groups, to just only the last number (instead of 16-28 I inserted just 28), regarding the y variable I attributed a number to each person's EVSs (Ecocentric-1, Technocentric-2 and Anthropocentric-3). I then applied these values to the *Pearson correlation coefficient calculator*[7], in order to find the r, which gives me the direction and strength of the correlation, bear in mind that a correlation to be strong needs to be above 0,7. The calculator gave me an r of 0. 2303, which can be considered weak, but not, non-existent.

This can be due to several factors the low distribution of responses through the age groups, or just because the correlation between these two variables is weak by nature. I am leaning towards the first option. Nevertheless, in order to prove this, more tests with a larger pool of samples for all the age groups, would be needed.

| | ***Anthropocentric*** | ***Ecocentric*** | ***Technocentric*** |
|---|---|---|---|
| ***16-28*** | *9* | *35* | *5* |
| ***29-44*** | *3* | *6* | *10* |
| ***45-60*** | *2* | *1* | *4* |
| ***61-79*** | *2* | *3* | *1* |
| ***80-97*** | - | *1* | - |

*Table 3 Answers for each EVSs devided by age group*

Nonetheless, the survey reached some interesting conclusion for example, the age group 16-28 is more ecocentric, however pending to the anthropocentrism and not to the technocentrism, as I thought that would happen, due to our dependence on technology. Moreover, a considerable amount of the responses in the age group 29-44 was ecocentric. The age group 61-79 is more ecocentric than anthropocentric, as breaking my hypothesis.

Furthermore, tendencies can be seen regarding some questions, in the first question:" Plastic products (straws, cups etc…) should be replaced by alternative materials less pollutant", for instance, the majority of people choose SA, about 44, and 41 choose A, thus only 5 people responded D, there is not response SD,

[7] https://www.socscistatistics.com/tests/pearson/default2.aspx

is also worthwhile noticing that, not surprisingly, the SA responses almost solely from the 16-28 age group.

The question, “The usage of fossil fuel cars should be reduced through legislation.” also offers some interesting responses, as a great part of this were D, an also considerable amount of people chose the option SD, and the majority of the A came from the age group 16-28. This is natural as the age groups, 29-44 and 45-60 are more reliant on cars, to go to their jobs, for instance.

The other questions do not show a clear tendency, other than differences in age group, but again the pool sample is too small to indicate that these divergences are a consequence of age.

**Conclusion**

The results above showed that my hypothesis was not entirely wrong, but not completely right either, as the age group 18-29 was mainly ecocentric and the 29-44 was mainly technocentric, nevertheless I thought that the EVSs in both the age groups would be more explicit. My hypothesis regarding the age group 45-60 was right, however, regarding the 61-79 group, my hypothesis was completely off, as they were more ecocentric than anthropocentric. Nevertheless, these, received very few answers, thus affirming these age groups mirror reality like the first two, would be incorrect. Furthermore, even though the 29-44 age group was the second with the more responses it doesn’t comes close to 16-28, therefore the results may be a little unprecise when compared to the 16-28 age group, meaning that, if more responses were received than the EVSs could be other.

The weak correlation of 0. 2303, showcases that the correlation between the age groups and the opinions regarding the strategies to stop global warming is weak but it is not impossible, hence perhaps a study with a larger and more balanced pool sample would arrive at the opposite conclusion. To portray the difference between opinions of the age groups, I used the EVSs, which although we could see, explicitly, to some extent, the presence of one specific EVS in a specific age group, 16-28, in the others this presence was not completely explicit,

may be due to the lack of a substantial quantity of response, were not so visible and explicit.

**Evaluation**

I believe that one of the biggest problems of this survey was the small sample size of some age groups, namely 45-60, 61-79 and 80-97, which made the results unreliable, when compared to the 16-28 and the 29-44. Furthermore, the discrepancy between the sample size between the age groups, made the analysis unbalanced, and I should have foreshadowed, in the beginning of the exploration, that this method of spreading the survey via *WhatsApp* would be more accessible to the younger generations (16-28) and should have thought of other ways to reach this older groups, asking the questions of the survey personally on the streets, for instance. Both factors may have influenced the correlation coefficient *r.*

Nevertheless, although unbalanced, the large sample pool in the first and second age group gave me a clear vision of the EVSs and particularities of both. Moreover, it also putted me under the impression that the weak corelation may not be due to both variables (age and opinion), but to the pool sample, as both the first two age groups convey, to some extent, explicit EVSs.

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